The Detectability Spectrum of Imaginary-Quadratic Invariants in Isomonodromy Data

(staged draft — author/affiliation withheld pending operator)

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Abstract

For the self-adjoint degree-two polynomial-continued-fraction (PCF) family with linear ODE $(ay')' - x^2 y = 0$, $a = \beta_2 x^2 + \beta_1 x + \beta_0$, we organise the question *which arithmetic invariants of the imaginary quadratic field $K = \mathbb{Q}(\sqrt{\Delta})$, $\Delta = \mathrm{disc}(a)$, leave a trace in the associated Painlevé/isomonodromy data* into a *detectability spectrum* indexed by (invariant $\times$ channel) pairs, with channels surface type, monodromy, and Stokes data. Our headline result is a Δ-*tower no-channel theorem*: under the standard Sakai/Okamoto premise that the surface type is a function of the local jet, the discrete Sakai surface type T is selected by β_0-free local data and is therefore constant on the locus $L = \{\beta_2, \beta_1, a_n \text{ fixed; } \beta_0 \text{ free; } \Delta \neq 0\}$, so T detects *no* non-trivial arithmetic invariant of K — not the class number h, the class group $\mathrm{Cl}(K)$, the unit order w, the set of ramified primes, nor the splitting type of any fixed prime. This strictly strengthens the previously recorded "CM field, not class number" observation from an empirical statement about one channel to a structural theorem about the discrete invariant. We make precise the distinction between *recoverable-in-principle* (R) and *intrinsic-channel-trace* (C) detectability, express the spectrum through a functor F from isomonodromy structures to arithmetic invariants, and characterise $\ker F$: trivial-as-a-channel for surface type (it forgets everything), and — for the richer connection channel — trivial in sense (R) but conjecturally equal to the L-value stratum $\{h, \mathrm{Cl}\}$ in sense (C). All symbolic and numerical claims are reproducible from a single self-contained script. We separate cleanly what is proven, verified, structural, and conjectural.

Contents

1 Introduction

Polynomial continued fractions (PCFs) attach to a recurrence $P_n = b_n P_{n-1} + a_n P_{n-2}$ a second-order linear ODE whose isomonodromic deformation carries a Sakai surface type, Stokes data at its irregular point, and a monodromy representation. A recurring empirical theme in this corpus is that the *class number* of the imaginary quadratic field naturally attached to such a family appears to be *invisible*: in the degree-two case, six $\Delta < 0$ families with class numbers spanning $h = 1$ and $h = 2$ exhibit identical Stokes ($S < 1$) and PSLQ-non-detection behaviour, evidence that "the controlling invariant is the sign of Δ ... and not the class number", sharpened by the pair in the *same* field $\mathbb{Q}(\sqrt{-7})$ at $h = 1$ versus $h = 2$ which identifies "the CM field, not the class number" as the structural carrier [1].

That observation is empirical and channel-specific. The present paper asks the structural question in full generality. We fix the self-adjoint degree-two family

$$(a(x)\,y')' - x^2 y = 0, \qquad a(x) = \beta_2 x^2 + \beta_1 x + \beta_0, \qquad b := a', \quad c := -x^2, \tag{1}$$

with $\Delta = \mathrm{disc}(a) = \beta_1^2 - 4\beta_2\beta_0$ and $K = \mathbb{Q}(\sqrt{\Delta})$, and we study *every* standard arithmetic invariant of K against *each* of three information channels. The organising object is a partition: each channel induces a partition of the parameter space, and an arithmetic invariant is detectable through a channel exactly when it is constant on the channel's fibres.

Contributions.

1. A *Δ-tower no-channel theorem* (Theorem 4): the discrete surface type T is constant on L, hence detects no non-trivial arithmetic invariant of K. The class-number case is the headline corollary; the proof is a symbolic computation that every type-selector is β_0-free, and is reproducible (§10, anchor module `nochannel/`).

2. A precise formalisation of the *(R) vs. (C)* distinction (§4): recoverable-in-principle versus intrinsic-channel-trace, the conceptual spine that explains how the connection channel can recover the field K yet leave the deep L-value invariants $\{h, \mathrm{Cl}\}$ with no known intrinsic reading.

3. A functor F from isomonodromy structures to arithmetic invariants and a characterisation of $\ker F$ per channel (§5).

4. A quick-results section (§6): the regulator is vacuous for imaginary quadratic K; unit-order detectability reduces to a finite reachability check of $\Delta \in \{-3,-4\}$; and the Δ-tower invisibility is essentially free from the main theorem.

5. Open problems (§7): the intrinsic h-trace in monodromy/Stokes data, with its conjectural link to $L(1, \chi_\Delta)$, and class-group structure detection.

Evidence discipline. We tag every assertion with an evidence class: [PROVEN] (a complete argument, given clearly-stated standard premises), [VERIFIED] (exact machine computation), [STRUCTURAL] (a precise statement supported by the structure but without a written proof here), [CONJECTURED], [UNKNOWN], and [VACUOUS] (vacuously constant, no content). A reconciliation table (§9) maps each numbered result to its `claims.jsonl` entry and flags any place where prose risks over-stating the recorded class.

2 Setup: channels, invariants, and the detectability map

2.1 The parameter space and the field

Fix $\beta_2 > 0$ and a numerator polynomial a_n (the PCF data; $a_n \equiv 1$ for the V_quad anchor, $a_n = n$ for the family QL15). The parameter space is

$$\mathcal{P} = \{p = (\beta_2, \beta_1, \beta_0) \in \mathbb{Q}^3 : \ \Delta(p) \neq 0\}, \qquad \Delta(p) = \beta_1^2 - 4\beta_2\beta_0.$$

Each p determines (1), its singular set $\{x_-, x_+, \infty\}$ with $x_\pm = (-\beta_1 \pm \sqrt{\Delta})/(2\beta_2)$, and the field $K(p) = \mathbb{Q}(\sqrt{\Delta(p)})$.

Remark 1 (discriminant normalisation [STRUCTURAL])**.** We write Δ for $\mathrm{disc}(a)$. The field invariant is the squarefree core: K depends only on the squarefree part of Δ, and $\mathrm{disc}(a)$ may differ from the *fundamental* discriminant d_K by a square. When Δ is itself a fundamental discriminant (e.g. $\Delta = -20$ for QL15) the order $\mathcal{O}_\Delta$ is the maximal order and the form class number coincides with h_K. In general we take $H(\Delta)$ to be the class number of the order $\mathcal{O}_\Delta$ (the number of $SL_2(\mathbb{Z})$-classes of primitive positive-definite forms of discriminant Δ); this is the invariant that varies along our loci.

2.2 Channels

A *channel* is a map $c\colon \mathcal{P} \to \mathsf{D}_c$ that factors through the isomonodromic/gauge equivalence appropriate to its data type. We consider three.

- **Surface type** $\mathsf{T}\colon \mathcal{P} \to \{\text{Sakai surface types}\}$: the discrete label of the space of initial conditions of the isomonodromic deformation of (1), in the sense of Sakai [2] and Okamoto.
- **Monodromy / connection** M: the full Riemann–Hilbert datum — the linear monodromy representation, the connection (Stokes-completed) matrices, and the *positions* $x_\pm$ of the singular points — modulo simultaneous conjugacy.
- **Stokes** S: the Stokes multipliers / constants at the rank-one irregular point at ∞ (for the V_quad anchor the relevant scalar is $S = 2\pi K$, the late-term resurgence constant).

These are listed in increasing fineness in the sense relevant below: T is a coarse discrete invariant, M retains continuous moduli, and S refines the local analytic data at ∞.

2.3 Arithmetic invariants and detectability

Let $\mathcal{K}$ be the set of imaginary quadratic fields. We consider the invariants

$$\Delta, \quad h = h_K, \quad \mathrm{Cl}(K), \quad w = \#\mathcal{O}_K^\times, \quad \mathrm{Ram}(K) = \{p : \ p \mid d_K\}, \quad \mathrm{Spl}_p(K) = \left(\tfrac{d_K}{p}\right), \quad R_K,$$

each a map $\iota\colon \mathcal{K} \to S_\iota$. Composing with $K\colon \mathcal{P} \to \mathcal{K}$ gives $\iota \circ K\colon \mathcal{P} \to S_\iota$.

Definition 2 (detectability through a channel)**.** An arithmetic invariant ι is *detectable through the channel* c if $\iota \circ K$ factors through c: there exists φ with $\iota \circ K = \varphi \circ c$ on $\mathcal{P}$; equivalently $\iota \circ K$ is constant on every fibre $c^{-1}(\delta)$. It is *invisible through* c if no such φ exists, i.e. some fibre of c carries at least two values of $\iota \circ K$.

The *detectability spectrum* is the table of (invariant, channel) pairs together with the status of Definition 2. Section 4 refines "factors through" into the (R)/(C) dichotomy.

3 The Δ-tower no-channel theorem

The engine is a symbolic fact about (1). All four parts are verified by exact computation in the anchor module (§10).

Lemma 3 (β_0-free local jet [PROVEN]/[VERIFIED])**.** *For* (1) *with* $\Delta \neq 0$*:*

(i) $b = a' = \beta_1 + 2\beta_2 x$ *and* $c = -x^2$ *are polynomial identities;* b *is* β_0*-free.*

(ii) At each finite root x_0 *of* a *the ODE* $y'' + (a'/a)y' + (c/a)y = 0$ *has a regular singular point with indicial polynomial* $r(r-1) + p_0 r + q_0 = r^2$*, where* $p_0 = \mathrm{Res}_{x_0}(a'/a) = 1$ *(because* $a'/a = \frac{d}{dx}\log a$*) and* $q_0 = \lim_{x\to x_0}(x-x_0)^2(c/a) = 0$*. Hence the local exponents are* $\{0,0\}$ *at both finite roots, independently of* β_0*; the resonance resolves to an* apparent *(no-logarithm) singularity, forced by* $b = a'$ *for all coefficients.*

(iii) At $x = \infty$*,* $x\,(a'/a) \to 2$ *(so* $a'/a \sim 2/x$*) and* $c/a \to -1/\beta_2$*; the point is irregular of Poincaré rank* 1 *with leading exponential rate* $\lambda = \pm 1/\sqrt{\beta_2}$ *(*$= \pm\theta_\infty$*,* $\theta_\infty = 2/\sqrt{\beta_2}$*), independent of* β_0*.*

(iv) $a(0) = \beta_0$*, so* $x = 0$ *is an ordinary point iff* $\beta_0 \neq 0$*.*

Consequently the entire type-selecting jet — pole orders, local/formal exponents, and apparent-singularity conditions — is independent of β_0*; the only* β_0*-dependence of the configuration is through the positions* $x_\pm = (-\beta_1 \pm \sqrt{\Delta})/(2\beta_2)$*.*

Proof (premise and computation). Parts (i)–(iv) are the exact symbolic computations of the anchor verifier (`nochannel/surface_type_nochannel_verify.py`, results recorded in `surface_type_nochannel_results.` The residue $p_0 = 1$ is the logarithmic-derivative identity; $q_0 = 0$ because c/a has only a simple pole at a simple root; the indicial polynomial is then r^2. At ∞ the stated limits give $\lambda^2 = 1/\beta_2$. The reduction "surface type is a function of this local jet" is the standard Sakai/Okamoto theory of spaces of initial conditions [2]; we take it as the (clearly stated) premise. □

Theorem 4 (Δ-tower no-channel [PROVEN], given the Sakai premise)**.** *Let* $L = \{\beta_2, \beta_1, a_n$ *fixed;* $\beta_0 \in \mathbb{Z};\ \Delta \neq 0\} \subset \mathcal{P}$*. Then the surface type is constant on* L*,*

$$\mathsf{T}(\beta_2, \beta_1, \beta_0, a_n) = \mathsf{T}(\beta_2, \beta_1, a_n) =: T_0,$$

while $H(\Delta(\beta_0))$ takes at least two distinct values on L. Consequently T *does not factor through H, and more strongly: for* every *arithmetic invariant ι of K that is non-constant on $\{K(p) : p \in L\}$ — in particular $\iota \in \{\Delta, h, \mathrm{Cl}, w, \mathrm{Ram}, \mathrm{Spl}_p\}$ — ι is* invisible through T*. The only arithmetic distinction* T *makes is the non-arithmetic confluence dichotomy $\Delta = 0$ vs. $\Delta \neq 0$.*

Proof. Constancy of T on L is Lemma 3. For the second clause it suffices that H is non-constant on L: by exact reduced-form count, on the worked locus $\beta_2 = 3, \beta_1 = 1$ ($\Delta = 1 - 12\beta_0$, all fundamental) one has $H(\beta_0{=}1{:}10) = 1, 3, 2, 5, 3, 7, 3, 8, 3, 10$, already two distinct values at $\beta_0 = 1, 2$. If $\iota \circ K = \varphi \circ \mathsf{T}$ held on L then $\iota \circ K$ would be constant on L (as T is), contradicting non-constancy of any such ι; for h this is immediate, and $\Delta, \mathrm{Cl}, w, \mathrm{Ram}, \mathrm{Spl}_p$ are non-constant along L as well (§6). The confluence remark is Lemma 3(iv) together with $\Delta = 0 \Leftrightarrow x_+ = x_-$. □

Corollary 5 (class number; the headline [PROVEN])**.** T *does not factor through the class number h. On the QL15 locus $\beta_2 = 3, \beta_1 = -2$ ($\Delta = 4 - 12\beta_0$, which passes through the QL15 field at $\beta_0 = 2$, $\Delta = -20 = \operatorname{disc} \mathbb{Q}(\sqrt{-5})$, $h = 2$) the type is constant while H takes the distinct values $\{1, 2, 3, 4, 6\}$ (these are order class numbers, some at non-fundamental Δ); the points $\beta_0 = 1$ ($K = \mathbb{Q}(\sqrt{-2})$, $\Delta = -8$, $h = 1$) and $\beta_0 = 2$ ($K = \mathbb{Q}(\sqrt{-5})$, $\Delta = -20$, $h = 2$) —* both fundamental*, so here h is the field class number h_K — lie on the same surface T_0 with $\Delta \neq 0$ and different h. The class-number conclusion thus rests only on fundamental discriminants; the non-fundamental orders enter no part of the argument.*

Remark 6 (what is, and is not, used [PROVEN])**.** The proof uses only (a) constancy of T (Lemma 3, symbolic) and (b) a finite computation that H takes ≥ 2 values. It does *not* use the identity of the constant surface T_0, nor any asymptotic growth of H; in particular it carries no exposure to the ineffectivity of the Siegel lower bound. The deposited determination $T_0 = \mathbb{Q}$-isomorphism class $D_5^{(1)}$ (Painlevé V, symmetry $W(A_3^{(1)})$, $\delta = -\frac{1}{2}$) is [VERIFIED] but lies *outside* the proof's critical path.

4 The (R)/(C) distinction

Definition 2 conflates two phenomena that Theorem 4 forces us to separate. The connection channel M retains the singular positions $x_\pm$, hence $\sqrt{\Delta}$, hence the field K; so *once K is recovered, every invariant of K is computable.* That is a weak, almost tautological sense of detection. The sharp content of a "spectrum" lives in a stronger sense.

Definition 7 (recoverable-in-principle, (R))**.** ι is *(R)-recoverable through c* if $\iota \circ K$ factors through c in the sense of Definition 2. Equivalently c refines the partition of $\mathcal{P}$ induced by $\iota \circ K$.

Definition 8 (intrinsic-channel-trace, (C))**.** Fix for each channel c an *admissible reading class Φ_c* of maps out of D_c: the features one is permitted to "read off". For M we take Φ_{M} to be the maps given by $\overline{\mathbb{Q}}$-algebraic functions of the entries of the monodromy/connection matrices and of the singular positions $x_\pm$ (the Riemann–Hilbert coordinates), *excluding* the arithmetic evaluation $K \mapsto \iota(K)$ itself. Then ι is *(C)-detectable through c* if $\iota \circ K = \varphi \circ c$ for some $\varphi \in \Phi_c$.

By construction (C) $\Rightarrow$ (R). The admissible class Φ_{M} formalises "read, do not solve": a (C)-trace is a bounded-complexity feature of the datum, not the result of recovering K and then invoking an oracle for $h(K)$. The boundary of Φ_{M} is a modelling choice; the conjectures below are robust to any reasonable choice because no reading of the monodromy datum is *known* to produce h other than via K.

Proposition 9 (field recovery [STRUCTURAL])**.** *K is (C)-detectable through* M*: the unordered pair $\{x_-, x_+\}$ of singular positions is part of the datum, and $\sqrt{\Delta} = \beta_2(x_+ - x_-)$ is an algebraic function of it, so $K = \mathbb{Q}(\sqrt{\Delta})$ is read off. Consequently* every *invariant of K is (R)-recoverable through* M*.*

This is [STRUCTURAL]: the map "RH datum $\mapsto \{x_\pm\} \mapsto \sqrt{\Delta}$" is forced by (1), but a fully written Riemann–Hilbert reconstruction theorem for this family is not part of the present corpus.

Remark 10 (the spectrum is a property of the pair, on one axis)**.** Combining Theorem 4 and Proposition 9: for T even (R) fails for every non-trivial ι (the strongest invisibility); for M, (R) holds for all ι, but (C) splits the invariants into a *geometric stratum* $\{\Delta, \mathrm{Ram}, \mathrm{Spl}_p, w\}$ that are algebraic functions of $x_\pm$ (hence (C)-detectable) and an *L-value stratum* $\{h, \mathrm{Cl}\}$ for which no admissible reading is known (§7).

5 The functor F and its kernel

Let $\mathcal{S}$ be the groupoid of isomonodromy structures of the family (1) (objects: ODEs on $\mathcal{P}$; isomorphisms: isomonodromic deformations and gauge equivalences). A channel c is a functor $\mathcal{S} \to \mathsf{D}_c$ to a discrete (for T) or moduli (for M) target; an invariant ι is a functor $\mathcal{S} \to S_\iota$ via K. Write F_c for the induced map on objects and define the *indistinguishability kernel* $\ker F_c = \{(p, p') : c(p) = c(p')\}$, the equivalence relation collapsed by c.

Proposition 11 (surface-type functor [PROVEN], given the Sakai premise)**.** *F_{T} is constant on each locus L (Theorem 4, under the standard Sakai/Okamoto premise that T is a function of the local jet), so $\ker F_{\mathsf{T}} \supseteq L \times L$. Hence the composite $\iota \circ K$ is constant on L for the partition induced by T, and the entire arithmetic spectrum $\{\Delta, h, \mathrm{Cl}, w, \mathrm{Ram}, \mathrm{Spl}_p\}$ lies in the part of $\mathcal{P}$ that F_{T} cannot separate. In words:* the discrete isomonodromy invariant is arithmetically blind — *it detects no invariant of K beyond the non-arithmetic dichotomy $\Delta = 0/{\neq}0$.*

Proposition 12 (connection functor [STRUCTURAL]/[CONJECTURED])**.** *F_{M} factors through K (Proposition 9). Therefore:*

- *the* provable *kernel is trivial in sense (R):* $\ker F_{\mathsf{M}}$ *separates any two parameters with $K(p) \neq K(p')$, so no invariant of K lies in* $\ker F_{\mathsf{M}}$ *(every invariant is a function of the recovered K); [STRUCTURAL];*
- *the* effective *kernel in sense (C) — the invariants with no admissible reading $\varphi \in \Phi_{\mathsf{M}}$ beyond "recover K, then evaluate" — contains the L-value stratum:* $\ker^{(C)} F_{\mathsf{M}} \supseteq \{h, \mathrm{Cl}\}$ *is the content of Conjecture 13. [CONJECTURED]*

Thus $\ker F$ has a clean two-tier description: for the discrete channel it swallows the whole arithmetic spectrum; for the connection channel it is trivial in (R) and conjecturally exactly the L-value stratum in (C). This is the strongest general statement the corpus currently supports.

6 Quick results

6.1 The regulator is vacuous [VACUOUS]/[PROVEN]

For every imaginary quadratic K the unit group has rank 0, so by the Dirichlet unit theorem the regulator is the empty determinant $R_K = 1$. A constant function carries no information:

"invisibility" of R_K is vacuous, not a no-channel phenomenon. Content appears only for *real* quadratic fields, where $R_K = \log \varepsilon_0$ varies; we note that QL15's field $\mathbb{Q}(\sqrt{-5})$ is imaginary, so this axis is dormant here and marks the natural next domain.

6.2 Ramification, splitting, and the geometric stratum [PROVEN](to T)/[STRUCTURAL](to M)

$\mathrm{Ram}(K) = \{p \mid d_K\}$ and $\mathrm{Spl}_p(K) = \left(\frac{d_K}{p}\right)$ are functions of Δ. By Theorem 4 they are invisible through T [PROVEN]. Through M they are (C)-detectable as algebraic functions of $\sqrt{\Delta}$ (Proposition 9) [STRUCTURAL]. We record the arithmetic nuance: the full splitting pattern over *all* primes equals the character χ_Δ, which determines Δ (a primitive real Dirichlet character determines its modulus), so all-prime splitting is equivalent to K; but finitely many small primes under-determine K.

6.3 Unit order and the reachability of $\Delta \in \{-3, -4\}$ [VERIFIED]

The unit order satisfies $w = 6 \iff \Delta = -3$, $w = 4 \iff \Delta = -4$, else $w = 2$; thus w is a function of Δ and is invisible through T [PROVEN]. Whether the extra-symmetry fields $\mathbb{Q}(i), \mathbb{Q}(\sqrt{-3})$ (with CM $j = 1728, 0$) are realised on a given locus is a finite check: on $a = x^2 + \beta_0$ one reaches $\Delta = -4$ at $\beta_0 = 1$; on $a = 3x^2 + 3x + \beta_0$ one reaches $\Delta = -3$ at $\beta_0 = 1$; but on the QL15 locus $a = 3x^2 - 2x + \beta_0$ one reaches *neither* for integer β_0, since $\Delta = -4$ requires $3\beta_0 - 1$ to be a perfect square, impossible as $3\beta_0 - 1 \equiv 2 \pmod 3$. [VERIFIED] (reproduce, §10). Whether $w > 2$ leaves an extra-symmetry trace in M is [STRUCTURAL].

7 Open problems

Conjecture 13 (no intrinsic h-trace [CONJECTURED])**.** *h is (C)-invisible through* M *and through* S*: there is no admissible reading $\varphi \in \Phi_{\mathsf{M}}$ (resp. Φ_{S}) with $h = \varphi \circ \mathsf{M}$ on the family. Equivalently, h is recoverable only via the field K, not as a bounded-complexity feature of the monodromy or Stokes datum.*

The heuristic is the class-number formula $h = \dfrac{w\sqrt{|d_K|}}{2\pi} L(1, \chi_\Delta)$: h is a *global* L-value, whereas Φ_{M} comprises local algebraic features. A first concrete probe is to sweep the Stokes constant $S = 2\pi K$ across the β_0-family and test for any correlation with h; currently only a single data point (S at the V_quad anchor) is computed, so the Stokes case is [UNKNOWN] rather than [CONJECTURED]. Proving Conjecture 13 (invisibility) would require a theorem that *no* algebraic function of the RH coordinates equals h along the family; exhibiting a trace (detectability) would require constructing the channel — e.g. relating an invariant trace field or a monodromy-group arithmetic invariant to $L(1, \chi_\Delta)$. Either direction needs machinery beyond the present corpus.

Problem 14 (class-group structure [UNKNOWN])**.** Detect $\mathrm{Cl}(K)$ (the group, not merely $h = \#\,\mathrm{Cl}$) through M or S. This is strictly harder than Conjecture 13: it asks for the group structure, and is recoverable in sense (R) via K but has no known intrinsic reading.

Problem 15 (real-quadratic extension [STRUCTURAL])**.** Repeat the analysis for parameter regimes with $\Delta > 0$, where the regulator acquires content ($R_K = \log \varepsilon_0$) and the unit group becomes infinite. The self-adjoint structure $b = a'$ persists, so Lemma 3 carries over and the surface-type invisibility should hold verbatim; the new question is whether the regulator or fundamental unit leaves a trace in M/S.

8 The spectrum, summarised

Invariant	vs. T	vs. M	vs. S
Δ (field K)	invisible [PROVEN]	detectable [STRUCTURAL]	moduli-dep. [STRUCTURAL]
Ram, Spl_p	invisible [PROVEN]	detectable [STRUCTURAL]	[UNKNOWN]
unit order w	invisible [PROVEN]	extra-symmetry [STRUCTURAL]	[UNKNOWN]
class number h	invisible [PROVEN]	(R) yes / (C) [CONJECTURED] no	[UNKNOWN]
class group Cl	invisible [PROVEN] (a fortiori)	(R) yes / (C) [UNKNOWN]	[UNKNOWN]
regulator R_K	vacuous [VACUOUS]	vacuous [VACUOUS]	vacuous [VACUOUS]

In the "vs. T" column, every [PROVEN] entry is [PROVEN] *modulo* the standard Sakai/Okamoto premise of Theorem 4 (surface type a function of the local jet); none is asserted unconditionally. The one-axis reading: a *geometric stratum* (Δ and its elementary functions) is invisible to the discrete type but faithful to K in the connection moduli; an *L-value stratum* (h, Cl) is invisible to the discrete type and conjecturally has no intrinsic trace anywhere; the regulator is vacuous for imaginary quadratic K.

9 Reconciliation with the claims ledger

Every numbered result maps to a recorded claim; we flag any prose that might be read above its recorded class.

Result	Ledger entry	Recorded class	Note
Lemma 3	`nochannel` NC-2..NC-5	[VERIFIED] (symbolic)	exact sympy
Theorem 4	`nochannel` NC-7	[PROVEN]*	$*$ given Sakai premise
Corollary 5	`nochannel` NC-6, DS-H-ST	[PROVEN]/[VERIFIED]	QL15 locus, $h(-20) = 2$
Prop. 9	DS-DISC-MON	[STRUCTURAL]	not a written RH theorem
Prop. 11	DS-FUNCTOR-T	[PROVEN]	from Thm 4
Prop. 12	DS-FUNCTOR-M	[STRUCTURAL]/[CONJECTURED]	(R) trivial, (C) conj.
§6 regulator	DS-REG-ALL	[VACUOUS]/[PROVEN]	$R_K \equiv 1$
§6 w reach.	DS-UNIT-MON	[VERIFIED]	finite check
Conj. 13	DS-H-MON, DS-H-STK	[CONJECTURED]/[UNKNOWN]	L-value heuristic only
Prob. 14	DS-CL-MON	[UNKNOWN]	harder than Conj. 13

Flags (prose vs. ledger).

- Theorem 4 is [PROVEN] *only modulo* the Sakai/Okamoto premise that T is a function of the local jet; the qualifier is stated at every occurrence of the result — abstract, contribution list (via back-reference to Theorem 4), the theorem and its proof, Proposition 11, and the summary table note. Do not cite Theorem 4 as unconditional.
- Proposition 9 ("M recovers K") is [STRUCTURAL], not [PROVEN]: it relies on a Riemann–Hilbert reconstruction that is asserted, not written. Any downstream use of "M detects Δ" inherits [STRUCTURAL].
- The phrase "M detects everything" (sense (R)) must never be softened into "M detects h intrinsically"; the latter is exactly Conjecture 13, recorded [CONJECTURED].
- The surface label $D_5^{(1)}$ is [VERIFIED] (deposited, MEDIUM band) and is kept *off* the critical path; the paper must not present it as first-principles [PROVEN].

10 Reproducibility

All symbolic and numerical claims are reproduced by a single self-contained script (CAS: `sympy`, exact symbolic for the ODE algebra; exact integer form-counts for class numbers, cross-checked against twelve classical anchors including $h(-20) = 2$, $h(-3) = h(-4) = 1$ with unit orders $6, 4$).

```
# foundational anchor (surface-type no-channel, Lemma 3.1 / Theorem 3.2):
python nochannel/surface_type_nochannel_verify.py
# detectability spectrum numerics (class numbers, splitting, reachability):
python files/detectability/detect_spectrum_verify.py
```

The anchor verifier exits 0 iff: the indicial polynomial is r^2 at both finite roots, $\lambda = \pm 1/\sqrt{\beta_2}$ at ∞ (all β_0-free); the QL15-locus class numbers are $\{1, 2, 3, 4, 6\}$ with $\Delta = -20, h = 2$ at $\beta_0 = 2$; and the prior-anchor sequence $1, 3, 2, 5, 3, 7, 3, 8, 3, 10$ is reproduced exactly.

Venue scan (editorial note, not for publication)

Three candidate venues consistent with the corpus history, *excluding* Experimental Mathematics (blacklisted) and *Notes on Number Theory and Discrete Mathematics* (blocked):

1. **SIGMA (Symmetry, Integrability and Geometry: Methods and Applications)** — open access; strong on Painlevé/isomonodromy and Sakai-surface classification; the surface-type framing fits squarely. Foreground the isomonodromy structure.
2. **The Ramanujan Journal** — continued fractions and arithmetic; the PCF/class-number framing and computer-assisted evidence are within scope. Foreground the no-channel theorem as a continued-fraction/arithmetic statement.
3. **Journal of Physics A: Mathematical and Theoretical** — isomonodromy, Stokes phenomena, Painlevé; the channel/Stokes framing and open problems fit. Foreground the detectability spectrum and the L-value open problem.

Fit caveats: SIGMA and J. Phys. A expect an integrable-systems narrative; the Ramanujan Journal expects the arithmetic angle foregrounded. The result is structural and conservative, which suits all three.

References

[1] *A CM predicate for degree-two polynomial continued fractions* (PCF-1), corpus manuscript `p12_pcf1_main.tex`; the empirical "sign of Δ / CM field, not the class number" statement and the $\mathbb{Q}(\sqrt{-7})$ pair (QL06, QL26).

[2] H. Sakai, *Rational surfaces associated with affine root systems and geometry of the Painlevé equations*, Comm. Math. Phys. **220** (2001), 165–229.

[3] K. Takeuchi, *Arithmetic triangle groups*, J. Math. Soc. Japan **29** (1977), 91–106.

[4] V_quad anchor, Zenodo `10.5281/zenodo.20455090` (Painlevé V on Sakai $D_5^{(1)}$; verdict VQ-N1).